MILITARY SPECIFICATION

SPATULA, DENTAL, No. 313

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a single-ended dental spatula suitable for spatulation of dental restorative materials, cements or medicaments.

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards, and drawing. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical, General Specifications.
MMM-A-250	Adhesive, Water-resistant, (For Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-sensitive Adhesive Paper, Water-Resistant (For Carton Sealing).

Military

MIL-P-116	Methods of Preservation.

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STANDARDS

Federal

FED-STD-151	Metal, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

22199	Spatula, Dental, No. 313.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18	Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM E 92	Standard Method of Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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3. REQUIREMENTS

3.1 General requirements. The spatula shall be in accordance with the applicable requirements of GG-I-526 and as specified herein.

3.2 Material. The spatula shall be fabricated from Class 4, type 440A martensitic stainless steel.

3.3 Hardness. After proper stress relieving, the Rockwell hardness of the operative end (blade) shall be C-52 to C-56. The shank and handle shall have a minimum Rockwell hardness of C-20.

3.4 Style, design and dimensions. The spatula shall be a single-ended dental spatula suitable for hand mixing (spatulation) of dental restorative materials, cements or medicaments. The spatula shall be a one-piece instrument and shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 22199 for style, design and dimensions. The spatula operative end (blade) shall be flexible.

3.5 Finish. The spatula (operative end, shank, and handle) shall be smooth and shall have a bright mirror finish.

3.6 Corrosion resistance. The spatula shall meet the requirements of GG-I-526 for resistance to copper sulfate and resistance to boil, when tested as specified in 4.4.3.1 and 4.4.3.2.

3.7 Identification marking. The handle of the spatula shall be permanently and legibly marked in accordance with the requirements of GG-I-526. In addition, the same surface shall be permanently and legibly marked with the instrument number "313".

3.8 Workmanship. Workmanship shall be first class throughout. The spatula shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw material, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the FSN, the Item Identification, the name of the manufacturer or supplier, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in Tables I and III (see 4.2.1 and 4.3) and Table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in Table I. The unit of product for examination purposes shall be one spatula.

Table I. Examination Criteria

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II, except that the acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one spatula.

Table II. Test Criteria

Characteristic	Reqmt.	Test procedure	Inspection level
Chemical composition	3.2	4.4.1	*
Hardness	3.3	4.4.2	S-2
Copper sulfate	3.6	4.4.3.1	S-2
Boil test	3.6	4.4.3.2	S-2

* A sufficient amount from each lot of raw materials used in the construction of the spatulas shall be selected for chemical analysis.

4.2.3 Sample unit usage. Where feasible, the same sample unit may be used for the determination of two or more test characteristics.

4.3 Examination. The spatula shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with, but not restricted to, the classification of defects listed in table III.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major A	
101	Style and design not as specified.
102	Finish on operative end not as specified.
103	Operative end not free from cracks, nicks, burrs, or unremoved creases.
104	Operative end contains rough, bent or ragged edges.
Major B	
151	Operative end dimensions not as specified.

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Table III. Classification of Defects (cont'd.)

Categories	Defects
Minor A	
201	Finish on shank and handle not as specified.
202	Finish not free from coarse grind marks or scratched surface areas.
203	Instrument not free from dirt, oil, polishing compound, or foreign matter.
204	Instrument number and the word "stainless" not present, legible, permanent, or properly located.
205	Spatula not permanently and legibly marked with name or trademark of the manufacturer.
Minor B	
251	Any dimension (other than those covered by 151) not as specified.

4.4 Test procedures.

4.4.1 Chemical composition. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151.

4.4.2 Hardness. Hardness shall be determined in accordance with ASTM Designation E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

4.4.3 Corrosion Resistance.

4.4.3.1 Copper sulfate test. The spatulas shall be subjected to the copper sulfate test as specified in GG-I-526.

4.4.3.2 Boil test. The spatulas shall be subjected to the boil test as specified in GG-I-526.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One spatula, as specified, constitutes one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive materials.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers a single-ended dental spatula suitable for spatulation of dental restorative materials, cements or medicaments.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required.
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Federal Stock Number (FSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog: 6520-720-9689, SPATULA, DENTAL, No. 313.

6.4 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1297

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-35793

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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